Detecting Impersonation Accounts on Instagram

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1 Objective (5 points)

The objective of this project is to identify potential fake or impersonation accounts on Instagram using a data-driven analysis of profile features. Motivated by observed instances of impersonation and an interest in cybersecurity, the study addresses a real-world online safety concern. Instagram was the best choice in this project due to its large user base and documented challenges with impersonation profiles. Using survey-collected profile data, the project applies similarity analysis to identify accounts with unusually high resemblance, which may indicate impersonation.

2 Outcome (5 point)

The final outcome of this project is a functional web-based impersonation detection tool that operationalizes the proposed similarity analysis framework. The tool features an interactive, survey-based that allows users to input publicly visible Instagram profile features. These inputs are processed using feature engineering and similarity-based statistical methods to identify accounts with high levels of resemblance. Results are displayed in an interactive dashboard that visualizes similarity scores, highlights highly similar account pairs, and flags profiles that may indicate potential impersonation behavior.

3 Impact (5 points)

This project empowers users to take an active role in identifying impersonation and maintaining greater control over their personal data, reducing reliance on platform's third party intervention. The resulting data provides value to social media researchers, brands, and cybersecurity teams by surfacing user-labeled signals of suspicious behavior and reducing manual investigation effort. From a data science perspective, the project generates high-quality, user-validated data that can be leveraged to improve impersonation detection models and reduce the cost of AI-based monitoring systems, which often struggle to identify accounts that have been created from the stolen personal information.

4 Literature Review (25 points)

1. The authors of Detection of Fake Instagram Accounts via Machine Learning Techniques investigated the impact of fake Instagram accounts on users and how per-

sonal data can be exploited for other purposes. They introduce profile and behavioral features used by machine learning models to detect fake accounts, providing empirical support for data-driven impersonation detection approaches.[5]

2. In An Efficient Method for Detection of Fake Accounts on the Instagram Platform, the authors propose a bagging classifier trained in labeled real and fake account data, achieving over 98% accuracy. This demonstrates the effectiveness of feature-driven ensemble-based approaches for identifying suspicious accounts.[4]
3. The study in Countering Social Media Cybercrime Using Deep Learning: Instagram Fake Accounts Detection presents a deep learning method using an LSTM network to detect fake Instagram accounts. Strong performance across public datasets shows that advanced neural models can effectively leverage profile features for impersonation detection.[3]
4. The authors in Machine Learning Interpretability to Detect Fake Accounts in Instagram emphasize model interpretability in detecting fake accounts. Using techniques such as SMOTE for class balancing and SHAP for feature explanation, the approach achieves approximately 96% accuracy and highlights key features distinguishing fake from real accounts.[1]
5. The authors of Instagram Fake Profile Detection Using An Ensemble Learning Method propose a hybrid machine learning framework combining ensemble classifiers (Random Forest and XGBoost) with data level techniques like SMOTE and hyperparameter tuning. Their results confirm that structured profile features and metadata can accurately detect fraudulent Instagram accounts, reinforcing the value of data-driven identification methods in social media security.[2]

5 Novelty (10 points)

Although existing studies primarily use supervised machine learning to detect fake Instagram accounts, this capstone offers several novel contributions. Most notably, it shifts from individual account classification to pairwise similarity analysis, allowing for the detection of impersonation clusters without pre-labeled datasets. By integrating interactive system validation through surveys, the project adds a layer of interpretability and trust that automated systems lack. Furthermore, the delivery of a web dashboard bridges the gap between high level academic research and a practical tool for every social media safety.

6 Data Source(s) (10 point)

1. InstaFake Dataset
   The dataset contains metric features capturing characteristics of Instagram accounts, including follower or following counts, biography length, presence of a profile picture, and account privacy settings. Widely used in research, it serves as a benchmark for detecting impersonated accounts and assessing the balance between fake and real accounts, making it a reliable resource for analyzing account authenticity.
2. Instagram Profile & Hashtag Dataset
   This dataset provides realistic Instagram profiles linked to specific hashtags, in-

cluding profile URLs, post counts, and engagement metrics. Collected via the professional Bright Data platform, it offers large-scale, realistic data essential for testing model performance on non-synthetic accounts and analyzing impersonation patterns in posts.

3. Fake Profile Detection Dataset
   This secondary dataset contains labeled social media accounts with numeric and categorical features. Although smaller, it is valuable for cross-validation, ensuring the models generalize well across diverse data sources and collection methods.

7 Approach (20 points)

1. **Background Research & Collecting Data:**
   Conduct background research to identify key features that differentiate realistic Instagram accounts from impersonated ones. Collect datasets and survey resources to support feature discovery and model validation.
2. **Survey Design:**
   Develop survey questions targeting the selected account attributes. This will provide human-validated signals to enhance model accuracy.
3. **Data Integration and Cleaning:**
   Consolidate datasets and survey responses into a unified format. Handle missing values, nulls, and duplicates to ensure clean, standardized data for analysis.
4. **Feature Engineering:**
   Create a similarity index by weighing specific profile attributes. Analyze features such as biography text, hashtags, follower/following ratios, post frequency, and engagement metrics to identify high overlaps indicative of impersonation.
5. **Model Development:**
   Apply pairwise similarity metrics to measure account overlap. Use clustering algorithms to group accounts exhibiting similar impersonation behaviors.
6. **UI Design:**
   Develop a web based dashboard, like Flask, to visualize similarity scores, flag high risk account pairs, and provide interpretable explanations for users. Include an interactive survey style interface for users to check potential impersonation of their own accounts.
7. **Validation and Final Report:**
   Validate the model using human-verified survey data. Document findings, generate a final report, and create posters to communicate the project's contributions to online safety.

Partial Success would be a validated similarity algorithm that accurately flags known duplicates in the dataset. Full success is the deployment of the Flask-based UI, where a user can input specific profile features and receive a real time similarity report, successfully matching the input against known impersonation patterns identified during the research phase.

8 Timeline (10 points)

- (2 Weeks) Data Collection & Background Research
- (1 Week) Data Preprocessing & Cleaning
- (2 Weeks) Exploratory Data Analysis & Feature Engineering
- (3 Weeks) Model Development
- (3 Weeks) Survey Interface Simulation & UI Design
- (1 Week) Report
- (1 Week) Poster & Presentation

9 Possible Challenges (10 points)

1. The model may mistakenly flag new accounts as impersonated. To address this, I will provide supporting evidence, such as follower or following ratios, post frequency, and similarity metrics.
2. High similarity does not always indicate impersonation (e.g., fan or business accounts), so I will implement a weighted feature analysis where unique identifiers like biography text or profile handles carry more weight than common metrics.
3. If tasks run behind schedule, I will reallocate time for quicker tasks.
4. All survey data will be anonymized, using only publicly visible metadata to ensure privacy and ethical compliance.

References

[1] Said Agoujil Anand Nayyar Amine Sallah, El Arbi Abdellaoui Alaoui. Machine learning interpretability to detect fake accounts in instagram. *ResearchGate*, 2022.

[2] Preeti Gulia Noha Alduaiji Piyush Kumar Shukla Shreyas J Bharti Goyal, Nasib Singh Gill. Instagram fake profile detection using an ensemble learning method. *Scientific Reports*, 15, 2025.

[3] Ghaida Alqarawi Murad A. Rassam Najla Alharbi, Bashayer Alkalifah. Countering social media cybercrime using deep learning: Instagram fake accounts detection. *Future Internet*, 16(10):367, 2024.

[4] Saeid Sheikhi. An efficient method for detection of fake accounts on the instagram platform. *Revue d'Intelligence Artificielle*, 34(4):429–436, 2020.

[5] Ioanna Roussaki1 Stefanos Chelas, George Routis. Detection of fake instagram accounts via machine learning techniques. *Computers*, 13(11):296, 2024.